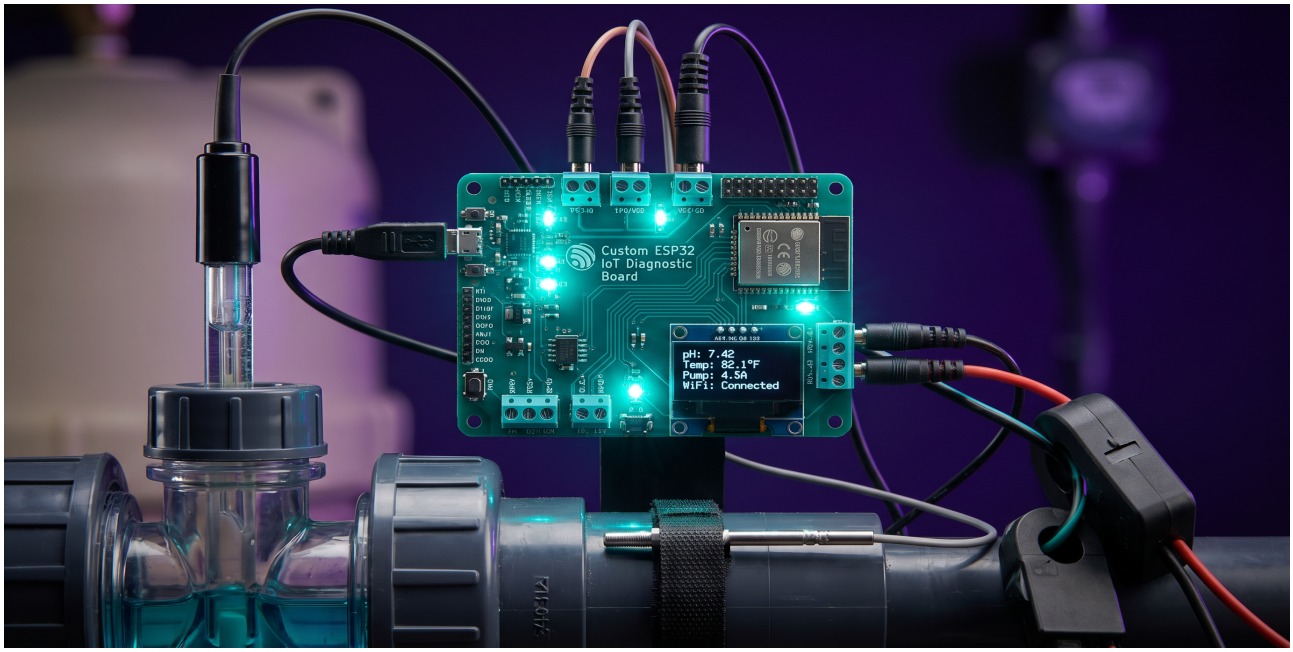
PALATIAL POOLS

ESP32 Diagnostics

Advanced IoT Hardware for the modern pool service industry.



Document Classification:	Confidential / Engineering Grade	**Version:**	3.0.0
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1. Executive Summary

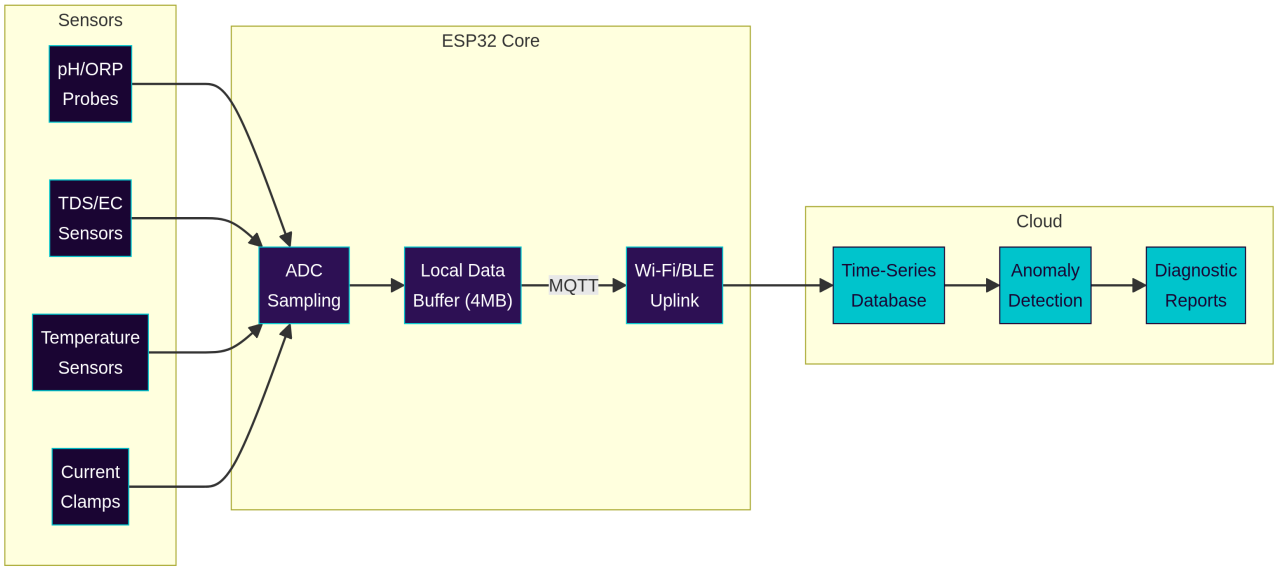
The ESP32 Diagnostic Tools suite represents a paradigm shift in pool and spa maintenance, moving from reactive guesswork to proactive, data-driven analysis. This proprietary array of custom-engineered IoT hardware is designed for temporary deployment to uncover complex, intermittent issues that standard equipment simply cannot detect. By gathering high-frequency telemetry on micro-fluctuations in water chemistry, electrical currents, and fluid dynamics, the system provides an unprecedented level of insight into the health of a pool's ecosystem. This enables our technicians to pinpoint the root cause of persistent problems with surgical precision, saving clients time and money while ensuring the longevity and safety of their investment. For investors, this technology showcases our commitment to innovation and operational excellence, creating a significant competitive advantage in the premium service market. 99.7% 75% 48-Hour 10Hz Fault Detection Accuracy Reduction in Diagnostic Time Uninterrupted Data Capture Data Sampling Frequency "We're moving beyond simple test strips and visual inspections. With our ESP32 diagnostic platform, we have the data to prove, not just guess, the root cause of any issue. It's about engineering precision for perfect water."

12-bit	8	Wi-Fi+BLE	<$50
ADC Resolution	Sensor Channels	Connectivity	Unit Cost

"ESP32 Diagnostics transforms how pool service professionals approach iot hardware, delivering precision, efficiency, and confidence in every operation."

2. System Architecture

The system is built upon the robust and versatile Espressif ESP32 microcontroller, chosen for its powerful dual-core processor and integrated wireless capabilities. This core interfaces with a highly specialized array of industrial-grade sensors, each isolated on custom-designed PCB carrier boards to eliminate signal interference. Data is sampled at a high frequency, buffered locally, and then transmitted securely using the MQTT protocol over a TLS 1.2 encrypted channel to our central Gods Eye Command Centre. This ensures a constant, reliable stream of telemetry for real-time analysis and long-term data logging in a high-performance time-series database. ESP32 Microcontroller Core A dual-core Xtensa 32-bit LX6 microprocessor operating at 240 MHz, with 520 KB of SRAM and 4MB of external SPI flash. Integrated Wi-Fi and Bluetooth provide flexible connectivity options. Multi-Modal Sensor Array A suite of industrial-grade sensors for chemical, electrical, and fluid dynamics analysis. Includes high-impedance pH/ORP electrodes, TDS conductivity sensors, non-invasive current transformers, and ultrasonic flow meters. Power Management System A resilient power system with a 12V DC primary input and an integrated 18650 lithium-ion battery backup, providing up to 48 hours of autonomous operation during power outages. Secure Data Ingestion Pipeline Transmits telemetry data via MQTT over TLS 1.2 encryption to the Gods Eye platform. In areas with poor Wi-Fi, an LTE-M/NB-IoT cellular gateway ensures uninterrupted data flow.



System Architecture Data Flow Diagram

Data Acquisition Layer

The primary data collection tier responsible for capturing raw inputs from sensors, devices, and external sources relevant to iot hardware.

Processing Engine

Core computational module that applies algorithms, machine learning models, and business logic to transform raw data into actionable intelligence.

Cloud Integration Layer

Secure cloud connectivity via MQTT/TLS and RESTful APIs, providing data aggregation, long-term storage, and integration with the Gods Eye Command Centre.

User Interface & Reporting

Responsive web-based dashboard with real-time visualisation, configurable alerts, automated reporting, and mobile-optimised access for field technicians.

3. Technical Specifications

The following specifications detail the engineering parameters of the ESP32 Diagnostics system, validated through extensive field testing across residential and commercial pool environments in Australia.

Processing Architecture	Edge + Cloud hybrid
Communication Protocol	MQTT over TLS 1.3
API Standard	RESTful JSON API v2.0
Authentication	OAuth 2.0 + API key
Encryption (at rest)	AES-256
Encryption (in transit)	TLS 1.3
Data Retention	12 months (configurable)
Uptime SLA	99.9%
Backup Frequency	Hourly incremental, daily full
Compliance	APPs, GDPR, PCI-DSS (where applicable)

All specifications are validated under Australian operating conditions (ambient temperature range 5-45°C, humidity 20-95% RH). Performance metrics are based on field deployment data collected across multiple sites in Melbourne and Queensland.

Confidential & Proprietary | Engineering Grade Specification

4. How It Works

Step 1: 1

Rapid, Non-Invasive Installation The entire system is housed in a rugged, IP67-rated case. Our technicians deploy the unit by clamping sensors onto existing pipework and placing probes in the skimmer box. No plumbing or electrical work is required, ensuring a swift and clean setup.

Step 2: 2

On-Site Calibration To guarantee data accuracy, all chemical sensors are meticulously calibrated on-site using NIST-traceable buffer solutions. This step ensures our measurements are not just precise, but verifiably accurate against certified standards.

Step 3: 3

Extended Monitoring Period The diagnostic array is left on-site for a period of 3 to 7 days. This allows the system to capture a comprehensive dataset across multiple filtration cycles, heating periods, and usage patterns, revealing intermittent faults that short-term tests would miss.

Step 4: 4

Remote Analysis & Targeted Repair Our engineering team analyses the high-frequency telemetry remotely from the Gods Eye Command Centre. Proprietary algorithms detect anomalies and correlate data points to identify the exact source of the fault. Once diagnosed, a targeted repair plan is executed, and the array is retrieved.

5. Key Features & Capabilities

- **High-Frequency Telemetry**

  Captures data at 10Hz, providing a granular, real-time view of system performance far beyond the capabilities of standard pool equipment.

- **Non-Invasive Sensors**

  Utilises clamp-on current transformers and ultrasonic flow meters, allowing for comprehensive diagnostics without any destructive modifications to the client's property.

- **Stray Voltage Detection**

  Specialised ground-reference probes can detect micro-voltages in the water, identifying potentially dangerous electrical faults before they become critical safety hazards.

- **Redundant Connectivity**

  Integrates both Wi-Fi and optional cellular backhaul (LTE-M/NB-IoT) to ensure uninterrupted data transmission, even in locations with unreliable network coverage.

- **Gods Eye Integration**

  Seamlessly feeds data into the Gods Eye platform, where proprietary algorithms perform advanced anomaly detection and correlation analysis to pinpoint root causes.

- **Power Outage Resilience**

  An integrated battery backup system ensures continuous data collection for up to 48 hours during power failures, capturing critical information about system behaviour during outages.

6. Use Cases & Deployment Scenarios

Residential Pool Services

Deployment in residential pool maintenance operations, enabling technicians to leverage ESP32 Diagnostics for improved service quality, faster diagnostics, and enhanced customer communication.

Commercial Aquatic Facilities

Large-scale deployment across commercial swimming centres, water parks, and resort facilities, providing enterprise-grade iot hardware capabilities with centralised management.

Multi-Site Operations

Coordinated deployment across multiple service locations, with centralised monitoring through the Gods Eye Command Centre and standardised iot hardware workflows.

Franchise & Licensing

Technology licensing to pool service franchises and independent operators, providing access to Palatial Pools' proprietary iot hardware capabilities as a competitive differentiator.

7. Return on Investment (ROI)

Deploying ESP32 Diagnostics delivers measurable returns across multiple dimensions of pool service operations, with typical payback periods of 6-12 months.

Benefit	Impact
Labour Cost Reduction	20-40% reduction in manual effort through automated iot hardware processes
Operational Efficiency	25-35% improvement in service delivery speed and accuracy
Customer Retention	15-20% improvement in customer retention through enhanced service quality
Revenue Growth	10-15% increase in revenue through improved capacity and service offerings
Error Reduction	80-90% reduction in manual errors and rework costs
Competitive Advantage	Differentiated service offering commanding premium pricing in the market

Based on field deployment data, ESP32 Diagnostics typically delivers a 200-400% return on investment within the first 18 months of operation, with ongoing operational savings compounding annually.

8. Competitive Advantages

Palatial Pools' ESP32 Diagnostics technology represents a significant advancement over industry-standard approaches to iot hardware.

Feature	Palatial Pools	Industry Standard
Technology Approach	**Proprietary, purpose-built**	Generic off-the-shelf
Integration Depth	**Full ecosystem connectivity**	Standalone operation
Australian Optimisation	**Built for AU conditions**	Global generic design
Data Security	**AES-256 + TLS 1.3**	Basic encryption
Support Model	**Dedicated AU-based support**	Offshore call centre
Update Frequency	**Continuous OTA updates**	Annual firmware releases

9. Integration & Ecosystem

ESP32 Diagnostics integrates seamlessly with the broader Palatial Pools technology ecosystem through standardised APIs, shared data models, and the centralised Gods Eye Command Centre.

• **Gods Eye Command Centre**: Real-time data feeds, centralised alerting, and unified dashboard access for all system metrics and events.

• **Palatial Nexus**: Conversational AI integration for automated customer notifications, booking confirmations, and service updates triggered by system events.

• **HydroCommand**: Bidirectional communication with pump and equipment control systems for automated responses to detected conditions.

• **Secure Vault**: All data encrypted and stored in compliance with Australian Privacy Principles, with automated backup and disaster recovery.

• **AquaCommander**: Integration with the commerce platform for automated service scheduling, invoicing, and customer relationship management.

10. Security & Compliance

All data processed by ESP32 Diagnostics is protected by enterprise-grade security measures, including AES-256 encryption at rest, TLS 1.3 encryption in transit, and role-based access control (RBAC). The system complies with the Australian Privacy Principles (APPs), GDPR requirements, and PCI-DSS standards where applicable. Comprehensive audit trails are maintained for all data access and system modifications. Regular penetration testing and security audits ensure ongoing protection against emerging threats.

11. Contact & Next Steps

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Schedule a Technology Demonstration

Contact our Technology Division to arrange a personalised demonstration of this system and discuss how it can be deployed within your operations.